

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of M. Sc. (Biotechnology/Microbiology/Biochemistry) Examination, April 2019

Course code & Name: MS 861 & Research Methodology

Date: 27/04/2019 Day: Saturday Time: 1:30 P.M. To 1:50 P.M. Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		10
1.	If you intend to work with <i>Francisella tularensis</i> , what will be the minimum level of laboratory biosafety containment in which you will perform your experiments?		01
	(i) BSL-1	(ii) BSL-3	
	(iii) BSL-2	(iv) BSL-4	
2.	The following is a conceptual structure within which research is conducted; it constitutes the blueprint for the collection, measurement and analysis of data.		01
	(i) Research Problem	(ii) Research Hypothesis	
	(iii) Research Design	(iv) Scientific Communication	
3.	If a researcher has little or no knowledge of the research area and s/he conducts small-scale feasibility or a pilot study before carrying out a detailed investigation, such type of research is known as _____.		01
	(i) Conceptual research	(ii) Longitudinal research	
	(iii) Exploratory research	(iv) Correlational research	
4.	The following is an example of secondary scientific literature.		01
	(i) Original research article	(ii) Review article	
	(iii) Conference proceedings	(iv) Rapid communication	

5.	The following test is used to determine the Operator Protection Factors (OPF) of biological safety cabinets.		01
	(i) Ames test	(ii) MTT assay	
	(iii) KI Discus test	(iv) Comet assay	
6.	An approval from Institutional Review Board (IRB) is required for research work on _____.		01
	(i) Rodents	(ii) Genetically modified microorganisms	
	(iii) Humans	(iv) Plant viruses	
7.	The use of someone else's concepts, hypothesis, conclusion, or interpretation without proper acknowledgement, in a way that implies to be the author's own is known as _____.		01
	(i) Plagiarism of text	(ii) Self-Plagiarism	
	(iii) Paraphrasing	(iv) Plagiarism of ideas	
8.	The following type of sample design is also known as chance or probability sampling where each and every item in the population has an equal chance of inclusion in the sampling.		01
	(i) Deliberate sampling	(ii) Quota sampling	
	(iii) Simple random sampling	(iv) Convenience sampling	
9.	The following is a reference management software.		01
	(i) Turnitin	(ii) OVID	
	(iii) Scopus	(iv) Mendeley	
10.	The following is NOT an abstracting or indexing bibliographic database.		01
	(i) MEDLINE	(ii) Scopus	
	(iii) EBSCO	(iv) Google Scholar	

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IV Semester of M. Sc. (Biotechnology/Microbiology/Biochemistry) Examination, April 2019

Course code & Name: MS 861 & Research Methodology

Date: 27/04/2019 Day: Saturday Time: 1:50 P.M. To 3:30 P.M. Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I		
Q - II Answer the following questions as directed (answer any two)		10
1.	Explain using a flow diagram the various steps of the research process.	05
2.	Explain the basic structure of a scientific research article.	05
3.	Give an overview of BSL-1 to 4 laboratory biosafety level containments.	05
SECTION – II		
Q-III Answer the following questions as directed (answer any five)		15
1.	What is an impact factor, how is it calculated? State its strengths and criticisms.	03
2.	Describe the features of primary, secondary and grey scientific literature.	03
3.	Describe the elements of a voluntary informed consent.	03
4.	Describe the composition of Institutional Animal Ethics Committee (IAEC).	03
5.	Explain the technique involved in defining a research problem.	03
6.	Explain different types of plagiarism in research publications.	03
7.	Describe the standards of a good research.	03